What changes an asteroid’s path?

antocciphysics

UNIT 2 / DAY 1 / HONORS

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Explain how gravity can change velocity without contact.

SUCCESS CRITERIA - I can...

1. Draw velocity tangent to a path and force toward the attracting body.
2. Distinguish motion from a change in motion.
3. State what a prediction of straight-line motion leaves out.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A simplified calculation predicts position (**-30**, **50**) km with bounds ±**3 km** in x and ±**5 km** in y. **Write a short briefing giving** both intervals and two reasons it is not a real asteroid collision forecast.

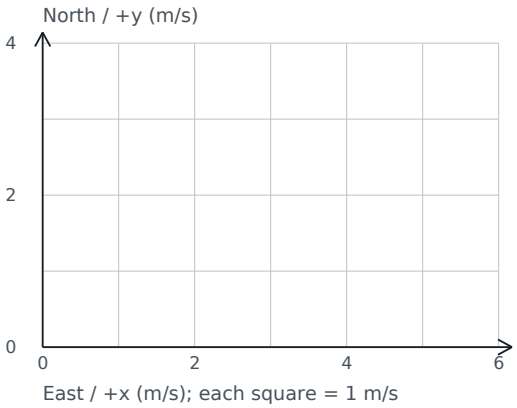
Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $a = F_{net}/m$.
Gravity attracts masses and can change speed, direction or both. Our asteroid scenario is fictional.

Worked reasoning: A fictional asteroid has a velocity arrow pointing right. A brief upward force gives it an upward change in velocity. The new velocity combines the original rightward motion with this change.

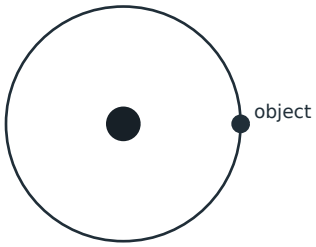
Your turn: A spacecraft is initially moving right. A short thruster burn adds an upward change in velocity.
1. **Draw the original velocity** and the change head-to-tail.
2. **Draw the final velocity** from the starting tail to the ending tip.



Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

1. **Sketch the velocity** and gravitational-force arrows at the marked point on the orbit.
2. Why do they point in different directions?

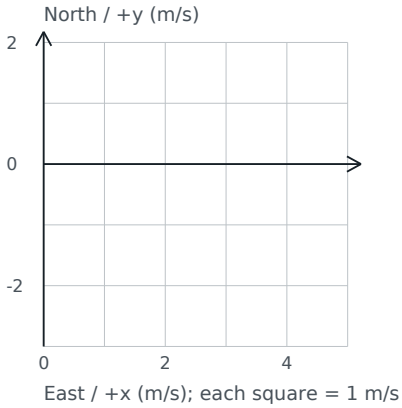


The center is the attracting body; the small dot on the circle is the orbiting object.

Marzano check - circle: ① ② ③

04 Apply the physics

A simulated probe starts with velocity (**4**, **0**) m/s. Its thruster produces acceleration (**0**, **-1**) m/s² for **2 s**. Plot the initial and final velocity vectors. **Find the final speed** and **explain whether** speed stayed constant.



Marzano check - circle: ① ② ③

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Continue down each column.

05 Explain your reasoning

A student says space has no gravity because astronauts float. **Explain how orbiting astronauts can** be falling.

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

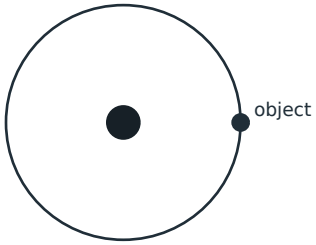
A **0.5 kg** system accelerates at **0.7 m/s²**. **Predict acceleration** for **0.9 kg** at the same net force. **Explain the effect of greater** resistance.
Use the table to keep the controlled and changed quantities distinct.

Trial	Mass	Net force	Acceleration
A	____	____	____
B	____	____	____

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07 Exit check

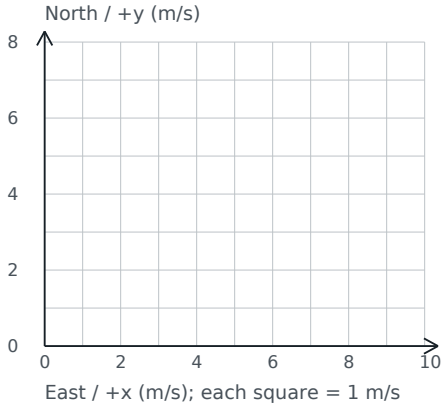
A force stays perpendicular to velocity during ideal circular motion. **Explain what changes** and what stays constant.



Marzano check - circle: ① ② ③

08 STRETCH: test the prediction

A simulated probe starts with velocity (**8**, **0**) m/s. Its thruster produces acceleration (**0**, **2**) m/s² for **3 s**.
1. **Draw the change in velocity** head-to-tail with the initial velocity.
2. **Find the final components** and speed.



Marzano check - circle: ① ② ③